FILE 8: questionnaire_for_repository.pdf

INVESTIGATING BLOCKCHAIN-BASED MODULAR CONSTRUCTION MANAGEMENT FOR AFFORDABLE HOUSING DELIVERY IN GHANA

COVER PAGE

Study Title: Blockchain-Based Modular Construction Management for Affordable Housing Delivery: Challenges and Implications for Housing Inequality in Emerging Cities

Research Institution: Hong Kong Polytechnic University (PolyU), Hong Kong

Principal Investigator: Peres Ofori
Co-Investigators: Albert CHAN Ping Chuen, Rotimi Abidoye, Derrick Alfa, Abdul Razak Ahmed

Purpose of Study: This questionnaire aims to collect primary data on stakeholder perceptions of barriers, benefits, and implications of blockchain-enabled modular construction for affordable housing delivery in Ghana. Your responses will help build an evidence-based framework for technology adoption in rapidly urbanising African cities.

Confidentiality: All responses are strictly confidential and anonymised. Data will be used solely for academic research.

Estimated Completion Time: 20–25 minutes

SECTION A: DEMOGRAPHIC AND PROFESSIONAL BACKGROUND

Instructions: Please tick (✓) or circle the appropriate response.

A1. Stakeholder Category

[ ] Construction Professional (Architect, Engineer, Quantity Surveyor, Project Manager)

[ ] Real Estate Developer

[ ] Modular Manufacturer / Supplier

[ ] Blockchain / Digital Construction Expert

[ ] Policy Maker / Regulator

[ ] Housing Beneficiary / Community Representative

A2. Gender

[ ] Male

[ ] Female

A3. Highest Educational Level

[ ] Tertiary

[ ] Secondary

[ ] Vocational

A4. Years of Professional Experience

[ ] <5 years

[ ] 5–10 years

[ ] 11–20 years

[ ] 20 years

A5. Study Location

[ ] Accra

[ ] Kumasi

SECTION B: AWARENESS AND KNOWLEDGE

Instructions: Please answer based on your knowledge and experience.

B1. Are you aware of modular construction methods?

[ ] Yes

[ ] No

B2. Are you aware of blockchain technology?

[ ] Yes

[ ] No

B3. How would you rate your knowledge of modular construction?
1 (Very Low) — 2 — 3 — 4 — 5 (Very High)

B4. How would you rate your knowledge of blockchain applications in construction?
1 (Very Low) — 2 — 3 — 4 — 5 (Very High)

B5. Has your organisation adopted modular construction?

[ ] Yes, regularly

[ ] Yes, occasionally

[ ] No

[ ] Not sure

SECTION C: BARRIERS TO BLOCKCHAIN-ENABLED MODULAR CONSTRUCTION ADOPTION

Instructions: Rate the significance of each barrier on a 5-point Likert scale (1 = not significant to 5 = extremely significant).

Code	Barrier Item	1	2	3	4	5
C1	High initial investment requirements	□	□	□	□	□
C2	Limited access to finance	□	□	□	□	□
C3	Uncertain return on investment	□	□	□	□	□
C4	Limited awareness and understanding	□	□	□	□	□
C5	Lack of skills and expertise	□	□	□	□	□
C6	Inadequate training and education	□	□	□	□	□
C7	Absence of clear policies and standards	□	□	□	□	□
C8	Complex approval processes	□	□	□	□	□
C9	Weak enforcement and incentives	□	□	□	□	□
C10	Design and integration complexities	□	□	□	□	□
C11	Limited technology and innovation	□	□	□	□	□
C12	Transportation and site constraints	□	□	□	□	□
C13	Perception of inferior design quality	□	□	□	□	□
C14	Limited design flexibility	□	□	□	□	□
C15	Concerns about appearance and finish	□	□	□	□	□
C16	Resistance to change	□	□	□	□	□
C17	Fragmented project delivery	□	□	□	□	□
C18	Coordination and supply chain challenges	□	□	□	□	□
C19	Limited industry experience and track record	□	□	□	□	□
C20	Unsuitable supply chain and infrastructure	□	□	□	□	□

Code	Barrier Item	1	2	3	4	5
C21	Lack of standardisation	□	□	□	□	□
C22	Risk aversion and scepticism	□	□	□	□	□
C23	Preference for traditional methods	□	□	□	□	□
C24	Low trust in new approaches	□	□	□	□	□
C25	Digital infrastructure limitations	□	□	□	□	□
C26	Power supply instability	□	□	□	□	□
C27	Fragmented industry structure	□	□	□	□	□
C28	Limited technical training	□	□	□	□	□
C29	Cultural resistance	□	□	□	□	□
C30	High import dependence	□	□	□	□	□

SECTION D: PERCEIVED BENEFITS AND IMPLICATIONS

Instructions: Rate your agreement on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree).

Code	Statement	1	2	3	4	5
D1	Modular construction can reduce construction costs.	□	□	□	□	□
D2	Modular construction can shorten project duration.	□	□	□	□	□
D3	Blockchain can improve supply chain transparency.	□	□	□	□	□
D4	Blockchain can enhance stakeholder collaboration.	□	□	□	□	□
D5	Blockchain can facilitate SME financing.	□	□	□	□	□
D6	Integrated adoption can improve housing affordability.	□	□	□	□	□

Code	Statement	1	2	3	4	5
D7	These technologies can reduce housing inequality.	□	□	□	□	□
D8	Government support is essential for adoption.	□	□	□	□	□

SECTION E: OPEN-ENDED QUESTIONS

E1. What are the most significant barriers to adopting blockchain-enabled modular construction in Ghana?

E2. What policy interventions would most effectively promote adoption?

E3. How can blockchain and modular construction together address housing inequality?

THANK YOU FOR YOUR PARTICIPATION

Your time and honest responses are invaluable for developing evidence-based policy for affordable housing delivery in Ghana and other rapidly urbanising African cities.

MAPPING TO RESEARCH OBJECTIVES AND HYPOTHESES

Questionnaire Section	Research Variable / Construct	Hypothesis(es) Tested	Research Question(s)
Section A	Demographics, professional background	Control	RQ1, RQ2
Section B	Awareness and knowledge	Descriptive	RQ1
Section C	Barriers to adoption (30 items)	H1, H2	RQ1
Section D	Perceived benefits and implications	H3	RQ2
Section E	Open-ended barriers and recommendations	Qualitative	RQ1, RQ2, RQ3

END OF QUESTIONNAIRE